

Search Query in FastAPI

What is a Search Query?

A search query is a way for users to filter data based on keywords or conditions.

Example:

When you type "john" in an admin dashboard to find a user → that "john" is the search query.

In APIs, it's sent as a query parameter in the URL:

GET /users?search=john

/users?search=john&page=2

Query String

- Starts with ?
- Key-value pairs: key=value
- Multiple parameters separated by &
- Handled in FastAPI using Query() from fastapi

How to Use Query Parameters in FastAPI

```
from fastapi import Query
```

```
def get_users(  
    search: str = Query(None, description="Search by username or email"),  
    page: int = Query(1, ge=1),  
    size: int = Query(10, gt=0, le=100)  
):
```

Where

search: str = Query(None) is Optional search term

% = wildcard (any number of any characters)

How Search Works in the Database (SQLAlchemy)

You used this code:

if search:

```
query = query.filter(
    (User.username.ilike(f"%{search}%")) |
    (User.email.ilike(f"%{search}%"))
)
```

if search:

Only apply the filter if the user provided a search term.

- If search = None or empty → skip
- Prevents unnecessary filtering

2. .filter()

Narrows down the database query.

- db.query(User) → gets all users
- .filter(condition) → returns only matching rows

Equivalent to SQL:

```
SELECT * FROM users WHERE username ILIKE '%ali%' OR email ILIKE '%ali%';
```

3. .ilike()

Case-insensitive pattern match (PostgreSQL, SQLite)

```
User.username.ilike("%pattern%")
```

% = wildcard (any number of any characters)

Users:

Search by username or email

[illegible]

Project

^

Cancel

Execute Clear

Responses

```
curl -X 'GET' \
'http://127.0.0.1:8000/project/?search-database&page=ilmo_records-5\'
-H 'accept: application/json'
-H "authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVC9i.ejY3PTEwZmQldmVlcnE6MkFkbGlzLWVmbGVudCkiOyBzaXIjcWtMTTBlN1llclA0JEVjeXBvdDgzcHVi.Y29zeWByMDVrSnlMcHRKRCkgPS1leUEmaWRo76c"
```

http://127.0.0.1:8000/project/?search-database&page=1&no_records=5

Code	Details
------	---------

```
{
  "page": 1,
  "records": 5,
  "total": 1,
  "pages": 1,
  "data": [
    {
      "id": 1,
      "name": "database",
      "desc": "all database-end work",
      "owner_id": 1
    }
  ]
}
```

```

{
  "page": 1,
  "records": 5,
  "total": 1,
  "pages": 1,
  "data": [
    {
      "id": 4,
      "name": "database",
      "des": "all database-end work",
      "owner_id": 1
    }
  ]
}

```

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